

4.5 Worksheet

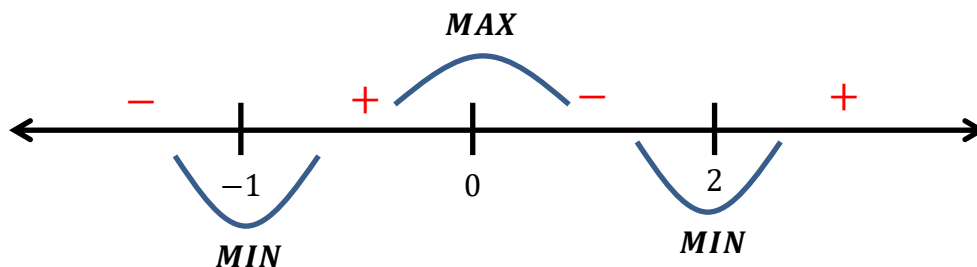
1. Find the intervals on which $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is increasing, decreasing, and local extrema.

We calculate: $f'(x) = 12x^3 - 12x^2 - 24x$.

$f'(x) = 0$ when

$$12x^3 - 12x^2 - 24x = 0 \rightarrow 12x(x^2 - x - 2) = 0 \rightarrow 12x(x-2)(x+1) = 0 \rightarrow x = -1, 0, 2$$

$f'(x)$ is never undefined.



Increasing/Decreasing

Increasing on: $(-1, 0) \cup (2, \infty)$

Decreasing on: $(-\infty, -1) \cup (0, 2)$

Local Minimums & Maximums

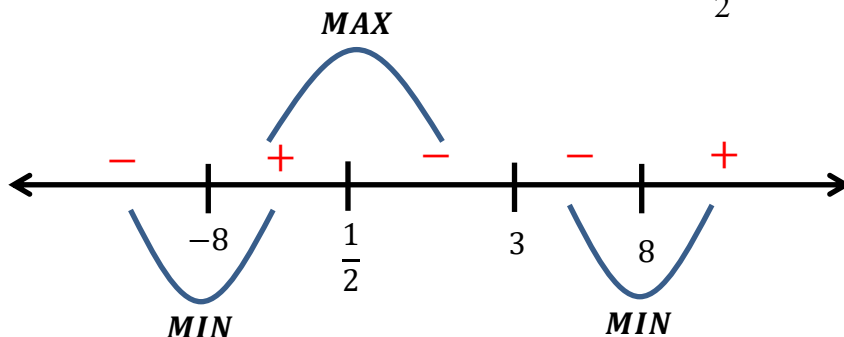
Local minimum: $f(-1) = 0$ & $f(2) = -27$

Local maximum: $f(0) = 5$

2. Suppose the function f has the first derivative of $f'(x) = e^{2x}(x^2 + 25)(x-3)^2(x^2 - 64)(2x-1)$.

List each interval upon which the function f is increasing and decreasing. State the value of x at which any local maximums or minimums of f occur.

The critical numbers are $x = 3$, $x = \pm 8$, & $x = \frac{1}{2}$



Increasing on: $\left(-8, \frac{1}{2}\right) \cup (8, \infty)$

Decreasing on: $(-\infty, -8) \cup \left(\frac{1}{2}, 3\right) \cup (3, 8)$

Local Maximum(s) occur(s) at: $x = \frac{1}{2}$

Local Minimum(s) occur(s) at: $x = \pm 8$

3. Find all intervals on which $f(x) = 3x^5 - 10x^4 + 10x^3 - 40x + 50$ is concave up and down. Find all inflection points on the graph of f .

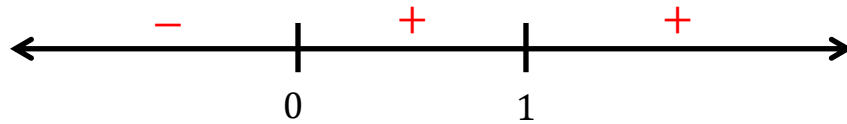
We calculate that: $f'(x) = 15x^4 - 40x^3 + 30x^2 - 40$

$$f''(x) = 60x^3 - 120x^2 + 60x$$

$f''(x) = 0$ when

$$60x^3 - 120x^2 + 60x = 0 \rightarrow 60x(x^2 - 2x + 1) = 0 \rightarrow 60x(x-1)^2 = 0 \rightarrow x = 0 \text{ \& } 1$$

$f'(x)$ is never undefined.



Concavity:

Concave up on: $(0,1) \cup (1,\infty)$

Concave down on: $(-\infty,0)$

Inflection Points:

$$x = 0, f(0) = 50 \rightarrow (0,50)$$

4. Find the critical numbers of $f(x) = (x-1)^2 \sqrt{x}$.

We calculate

$$f'(x) = 2(x-1)\sqrt{x} + (x-1)^2 \frac{1}{2\sqrt{x}}$$

$$= \frac{1}{2\sqrt{x}}(x-1)[4x + (x-1)]$$

$$= \frac{1}{2\sqrt{x}}(x-1)[5x-1]$$

$f'(x) = 0$ when

$$\frac{1}{2\sqrt{x}}(x-1)[5x-1] = 0 \rightarrow x = 1 \text{ \& } \frac{1}{5}$$

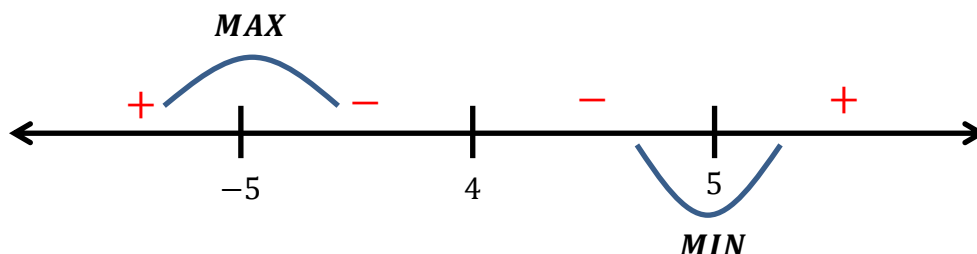
$f'(x)$ is undefined when $x = 0$.

The critical numbers are therefore $x = 0$, $x = 1$, & $x = \frac{1}{5}$

5. Suppose that $g(x)$ is a function whose first derivative is $g'(x) = \frac{8e^x(x^2+36)(x^2-25)(x-4)^{10}}{\ln(x^2+9)}$.

Find all critical numbers of $g(x)$. At each critical number, state whether $g(x)$ has a local maximum, local minimum or neither at that point.

The critical numbers are $x = \pm 5$ & $x = 4$



Increasing on: $(-\infty, -5) \cup (5, \infty)$

Decreasing on: $(-5, 4) \cup (4, 5)$

Local Maximum(s) occur(s) at: $x = -5$

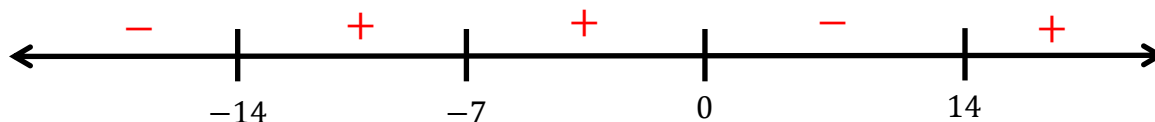
Local Minimum(s) occur(s) at: $x = 5$

6. A polynomial $f(x)$ has the second derivative $f''(x) = xe^x(x+7)^6(x^2-196)(x^2+25)$. State the interval where f is concave up and concave down. State each x -value at which the graph of $f(x)$ has an inflection point.

$f''(x) = 0$ when

$x = 0, x = -7, x = -14, \text{ \& } x = 14$

$f''(x)$ is never undefined.



Concavity:

Concave up on: $(-14, -7) \cup (-7, 0) \cup (14, \infty)$

Concave down on: $(-\infty, -14) \cup (0, 14)$

Inflection Points:

Occur at $x = -14, x = 0, \text{ \& } x = 14$

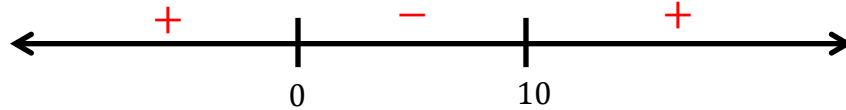
7. Find each interval of concavity and each inflection point for the function $f(x) = \frac{x-5}{x^3}$.

We calculate: $f'(x) = \frac{x^3(1) - 3x^2(x-5)}{x^6} = \frac{x^3 - 3x^3 + 15x^2}{x^6} = \frac{15x^2 - 2x^3}{x^6} = \frac{15 - 2x}{x^4}$

Therefore, $f''(x) = \frac{x^4(-2) - 4x^3(15-2x)}{x^8} = \frac{-2x^4 - 60x^3 + 8x^4}{x^8} = \frac{6x^4 - 60x^3}{x^8} = \frac{6x - 60}{x^5} = \frac{6(x-10)}{x^5}$

$f''(x) = 0$ when $x = 10$.

$f''(x)$ is undefined when $x = 0$.



Concavity:

Concave up on: $(-\infty, 0) \cup (10, \infty)$

Concave down on: $(0, 10)$

Inflection Points:

Occur at $x = 10$, $f(10) = \frac{5}{1000} = \frac{1}{200}$.

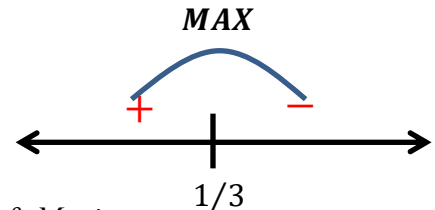
So we have the point: $\left(10, \frac{1}{200}\right)$

Note that $x = 0$ is not in the domain of f so it cannot be a point of inflection.

8. Find when the function $f(x) = 8 + 5xe^{-3x}$ is increasing, decreasing, concave up, and concave down. Also find all relative extrema and inflection points of $f(x)$.

We calculate: $f'(x) = 5e^{-3x} - 15xe^{-3x} = 5e^{-3x}(1-3x)$

$f'(x) = 0$ when $x = \frac{1}{3}$. $f'(x)$ is never undefined.



Increasing/Decreasing

Increasing on: $\left(-\infty, \frac{1}{3}\right)$

Decreasing on: $\left(\frac{1}{3}, \infty\right)$

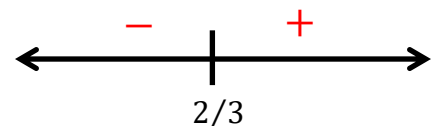
Local Minimums & Maximums

Local minimum: None

Local maximum: $f\left(\frac{1}{3}\right) = 8 + \frac{5}{3e}$

We calculate: $f''(x) = -15e^{-3x}(1-3x) - 15e^{-3x} = -15e^{-3x}[(1-3x)+1] = -15e^{-3x}(2-3x)$

$f''(x) = 0$ when $x = \frac{2}{3}$. $f''(x)$ is never undefined.



Concavity:

Concave up on: $\left(\frac{2}{3}, \infty\right)$

Concave down on: $\left(-\infty, \frac{2}{3}\right)$

Inflection Points:

$x = \frac{2}{3}$, $f\left(\frac{2}{3}\right) = 8 + \frac{10}{3e^2}$ $\left(\frac{2}{3}, 8 + \frac{10}{3e^2}\right)$

9. A bullet is fired upwards from the ground at an initial velocity of 1200 feet per second. The height s (in feet) of the bullet above the ground after t seconds is $s = 1200t - 6t^2$ on Mars and $s = 1200t - 16t^2$ on Earth. How much higher will the bullet travel on Mars than on Earth? (Hint: What is the slope of the tangent line when the bullet is at it highest point?)

Height on Mars: $v(t) = s'(t) = 1200 - 12t$

The maximum height is reached when the velocity is zero.

$$0 = 1200 - 12t$$

$$t = 100 \text{ sec}$$

We compute the height at this time: $s(100) = 1200(100) - 6(100)^2 = 60,000 \text{ ft}$

Height on Earth: $v(t) = s'(t) = 1200 - 32t$

The maximum height is reached when the velocity is zero.

$$0 = 1200 - 32t$$

$$t = 37.5 \text{ sec}$$

We compute the height at this time: $s(37.5) = 1200(37.5) - 16(37.5)^2 = 22,500 \text{ ft}$

Note that $60,000 - 22,500 = 37,500 \text{ ft}$

Therefore the bullet will travel $37,500 \text{ ft}$ higher on Mars than on Earth.

10. A rock is thrown upward from the surface of the moon. Between the time that the rock is thrown and the time that the rock hits the ground, the rock's height is given by the formula $s(t) = 24t - 0.8t^2$, where t is the number of seconds since the rock is first thrown and $s(t)$ is measured in meters above the moon's surface.

- a. Find a formula for the rock's velocity at time t .

$$v(t) = s'(t) = 24 - 1.6t$$

- b. Find a formula for the rock's acceleration at time t .

$$a(t) = v'(t) = -1.6$$

- c. How fast was the rock moving, and in which direction, at the end of 25 seconds.

$$v(25) = 24 - 1.6(25) = -16$$

The rock is moving downward at a velocity of 16 m/s at the end of 25 seconds.

- d. What is rock's maximum height and at what time does it reach that height?

When the rock obtains its maximum height its velocity is 0 m/s .

$$0 = 24 - 1.6t$$

$$t = 15$$

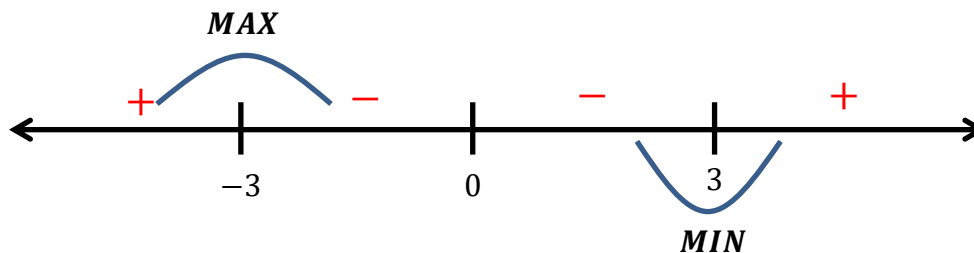
$$s(15) = 24(15) - 0.8(15)^2 = 180$$

The rock reaches a maximum height of 180 m after 15 seconds.

11. Sketch a graph of the curve $f(x) = x^5 - 15x^3$.

We calculate: $f'(x) = 5x^4 - 45x^2 = 5x^2(x^2 - 9)$

$f'(x) = 0$ when $x = 0, -3, 3$. $f'(x)$ is never undefined.



Increasing/Decreasing

Increasing on: $(-\infty, -3) \cup (3, \infty)$

Decreasing on: $(-3, 0) \cup (0, 3)$

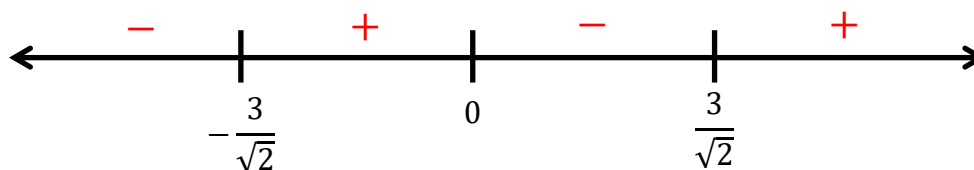
Local Minimums & Maximums

Local minimum: $f(3) = -162$

Local maximum: $f(-3) = 162$

We calculate: $f''(x) = 20x^3 - 90x = 10x(2x^2 - 9)$

$f''(x) = 0$ when $x = 0, \frac{3}{\sqrt{2}}, -\frac{3}{\sqrt{2}}$. $f''(x)$ is never undefined.



Concavity:

Concave up on: $\left(-\frac{3}{\sqrt{2}}, 0\right) \cup \left(\frac{3}{\sqrt{2}}, \infty\right)$

Concave down on: $\left(-\infty, -\frac{3}{\sqrt{2}}\right) \cup \left(0, \frac{3}{\sqrt{2}}\right)$

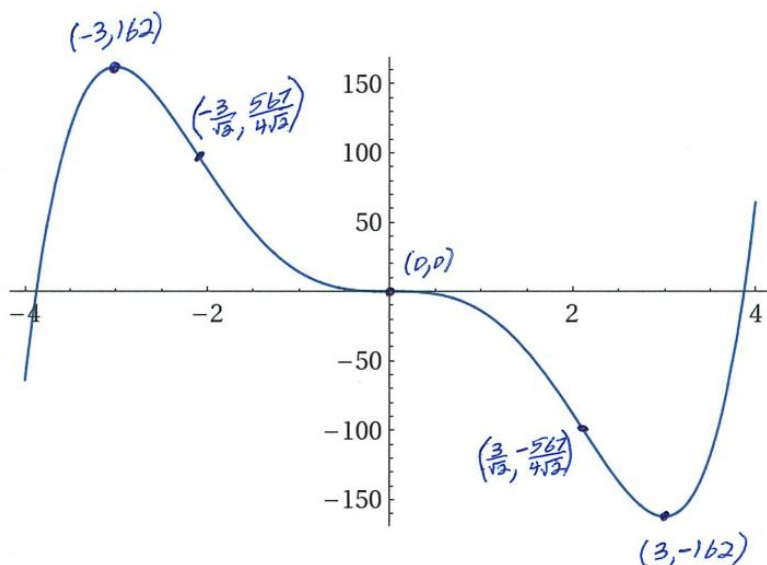
Inflection Points:

$x = -\frac{3}{\sqrt{2}}, f\left(-\frac{3}{\sqrt{2}}\right) = \frac{567}{4\sqrt{2}}$

$\left(-\frac{3}{\sqrt{2}}, \frac{567}{4\sqrt{2}}\right)$

$x = \frac{3}{\sqrt{2}}, f\left(\frac{3}{\sqrt{2}}\right) = -\frac{567}{4\sqrt{2}}$

$\left(\frac{3}{\sqrt{2}}, -\frac{567}{4\sqrt{2}}\right)$



12. Find the absolute extrema of $f(x) = \frac{1}{x^2 - x}$ on the interval $(0,1)$.

We are not looking at a closed interval so we cannot be sure $f(x)$ has an absolute max. & min. on $(0,1)$.

We calculate: $f'(x) = \frac{-2x+1}{(x^2-x)^2}$.

Notice that $f'(x)$ is undefined when $x=0$ or $x=1$.

Notice that $f'(x)=0$ when $x=\frac{1}{2}$.

Therefore the only critical numbers of the function $f(x)$ in the interval $(0,1)$ is $x=\frac{1}{2}$.

We also consider the limits as we approach the endpoints of this open interval.

$$f\left(\frac{1}{2}\right) = -4$$

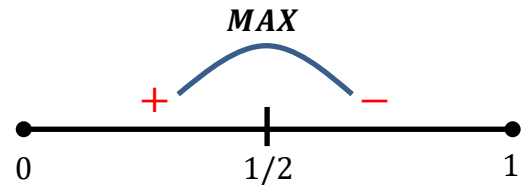
$$\lim_{x \rightarrow 0^+} \frac{1}{x^2 - x} = \lim_{x \rightarrow 0^+} \frac{1}{x(x-1)} \rightarrow \frac{1}{0^+ \cdot 0^-} \rightarrow -\infty$$

$$\lim_{x \rightarrow 1^-} \frac{1}{x^2 - x} = \lim_{x \rightarrow 1^-} \frac{1}{x(x-1)} \rightarrow \frac{1}{1 \cdot 0^-} \rightarrow -\infty$$

Can avoid by considering the number line

Therefore we see that $f(x)$ has:

- NO absolute minimum
- an absolute maximum $f\left(\frac{1}{2}\right) = -4$



13. Find the absolute extrema of $f(x) = e^{(x^3-3x^2)}$ on the interval $(0,\infty)$.

We are not looking at a closed interval so we cannot be sure $f(x)$ has an absolute max. & min. on $(0,\infty)$.

We calculate: $f'(x) = e^{(x^3-3x^2)}(3x^2-6x) = 3x(x-2)e^{(x^3-3x^2)}$

Notice that $f'(x)$ is never undefined.

Notice that $f'(x)=0$ when $x=0$ & $x=2$.

Therefore the only critical numbers of the function $f(x)$ in the interval $(0,\infty)$ is $x=2$.

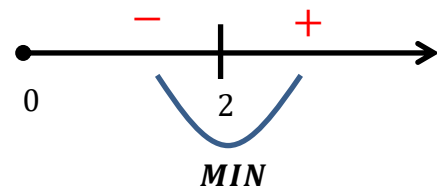
We also consider the limits as we approach the endpoints of this open interval.

$$f(2) = e^{-4}$$

$$\lim_{x \rightarrow 0^+} e^{(x^3-3x^2)} = e^0 = 1$$

$$\lim_{x \rightarrow \infty} e^{(x^3-3x^2)} = \infty$$

Can avoid by considering the number line



Therefore we see that $f(x)$ has:

- absolute minimum $f(2) = e^{-4}$
- NO absolute maximum

14. Determine the absolute minimum y-value on the graph of $y = 2e^{4x} - 648x + 38$. Simplify your answer fully.

We calculate that $f'(x) = 8e^{4x} - 648 = 8(e^{4x} - 81)$.

Notice that $f'(x)$ is never undefined.

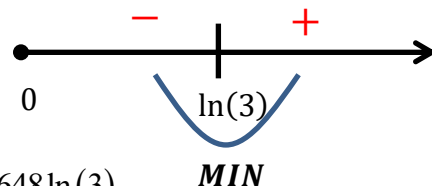
Notice that $f'(x) = 0$ when:

$$8(e^{4x} - 81) = 0 \rightarrow e^{4x} - 81 = 0 \rightarrow e^{4x} = 81 \rightarrow 4x = \ln(81) \rightarrow x = \frac{1}{4}\ln(81) = \ln(3)$$

Therefore the critical numbers of the function $f(x)$ are $x = \ln(3)$.

$$f(\ln(3)) = 162 - 648\ln(3) + 38 = 200 - 648\ln(3)$$

Therefore we see that $f(x)$ has an absolute minimum $f(\ln(3)) = 200 - 648\ln(3)$.



15. Determine the x-value at which the function $f(x) = \frac{e^x}{e^{2x} + 5}$ attains its absolute maximum value.

$$\text{We calculate that } f'(x) = \frac{(e^{2x} + 5)e^x - e^x e^{2x} (2)}{(e^{2x} + 5)^2} = \frac{e^x(e^{2x} + 5 - 2e^{2x})}{(e^{2x} + 5)^2} = \frac{e^x(5 - e^{2x})}{(e^{2x} + 5)^2}.$$

Notice that $f'(x)$ is never undefined.

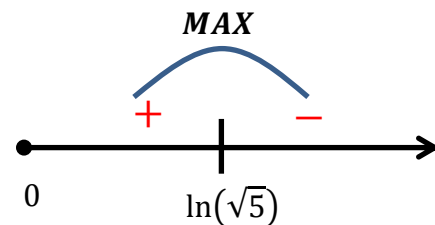
Notice that $f'(x) = 0$ when:

$$5 - e^{2x} = 0 \rightarrow e^{2x} = 5 \rightarrow 2x = \ln(5) \rightarrow x = \frac{1}{2}\ln(5) = \ln(\sqrt{5})$$

Therefore the critical numbers of the function $f(x)$ are $x = \ln(\sqrt{5})$.

$$f(\ln(\sqrt{5})) = \frac{\sqrt{5}}{5+5} = \frac{\sqrt{5}}{10}$$

Therefore we see that $f(x)$ has an absolute maximum $f(\ln(\sqrt{5})) = \frac{\sqrt{5}}{10}$



16. What are the coordinates (x, y) for the highest point on the graph of the function $g(x) = 10xe^{-2x}$.

We calculate $g'(x) = 10e^{-2x} + 10xe^{-2x}(-2) = 10e^{-2x}(1-2x)$.

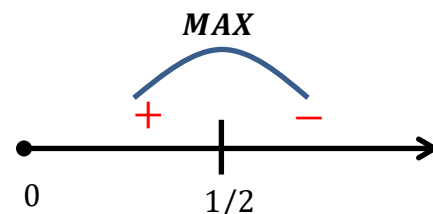
Notice that $f'(x)$ is never undefined.

Notice that $f'(x) = 0$ when $1-2x = 0 \rightarrow x = \frac{1}{2}$

Therefore the critical numbers of the function $f(x)$ are $x = \frac{1}{2}$.

$$f\left(\frac{1}{2}\right) = \frac{5}{e}$$

Therefore we see that $f(x)$ has an absolute maximum $f\left(\frac{1}{2}\right) = \frac{5}{e}$.

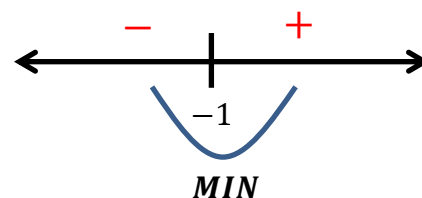


17. Find the intervals where $f(x) = xe^x$ is increasing, decreasing, concave up, and concave down. Find all local extrema and inflection points of f .

First Derivative: $f'(x) = e^x + xe^x$

$f'(x) = 0$ when $e^x + xe^x = 0 \rightarrow e^x(1+x) = 0 \rightarrow x = -1$

$f'(x)$ is never undefined.



Increasing/Decreasing

Increasing on: $(-1, \infty)$

Decreasing on: $(-\infty, -1)$

Local Extrema

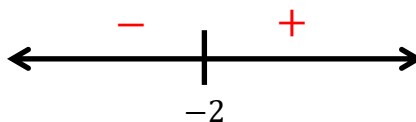
Local minimum: $f(-1) = -\frac{1}{e}$

Local maximum: None

Second Derivative: $f''(x) = e^x + e^x + xe^x = 2e^x + xe^x$

$f''(x) = 0$ when $2e^x + xe^x = 0 \rightarrow e^x(2+x) = 0 \rightarrow x = -2$

$f''(x)$ is never undefined.



Concavity:

Concave up on: $(-2, \infty)$

Concave down on: $(-\infty, -2)$

Inflection Points:

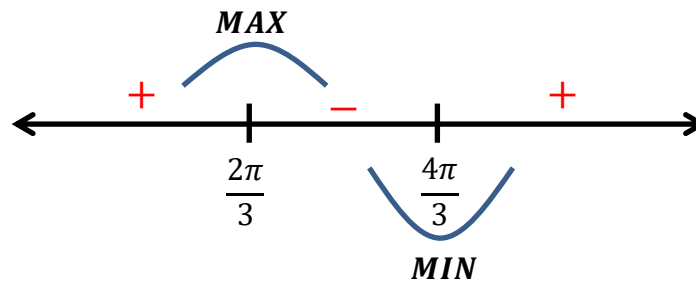
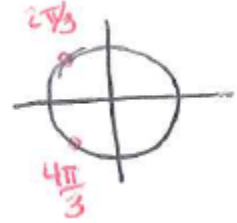
$x = -2, f(-2) = -\frac{2}{e^2} \quad \left(-2, -\frac{2}{e^2}\right)$

18. Find the intervals where $f(x) = x + 2\sin(x)$ is increasing, decreasing, concave up and concave down on the interval $[0, 2\pi]$. Find all the local extrema and inflection points of f .

First Derivative: $f'(x) = 1 + 2\cos(x)$

$$f'(x) = 0 \text{ when } 1 + 2\cos(x) = 0 \rightarrow \cos(x) = -\frac{1}{2} \rightarrow x = \frac{2\pi}{3} \text{ or } x = \frac{4\pi}{3}$$

$f'(x)$ is never undefined.



Increasing/Decreasing

Increasing on: $\left(0, \frac{2\pi}{3}\right) \cup \left(\frac{4\pi}{3}, 2\pi\right)$

Decreasing on: $\left(\frac{2\pi}{3}, \frac{4\pi}{3}\right)$

Local Minimums & Maximums

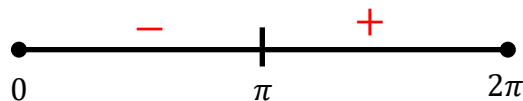
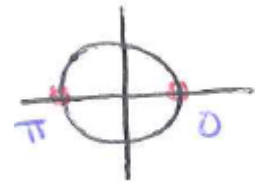
Local minimum: $f\left(\frac{4\pi}{3}\right) = \frac{4\pi}{3} - \sqrt{3}$

Local maximum: $f\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} + \sqrt{2}$

Second Derivative: $f''(x) = -2\sin(x)$

$$f''(x) = 0 \text{ when } -2\sin(x) = 0 \rightarrow \sin(x) = 0 \rightarrow x = 0 \text{ or } x = \pi \text{ or } x = 2\pi$$

$f''(x)$ is never undefined.



Concavity:

Concave up on: $(\pi, 2\pi)$

Concave down on: $(0, \pi)$

Inflection Points:

$x = \pi, f(\pi) = \pi$ (π, π)